Resource Theories

A mathematical framework: abstract and quantum

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Contents

0 Abstract Framework **2**
- 0.1 Resource theories as symmetric monoidal categories . . . 2
- 0.2 Partitioned process theories . . . 2
- 0.3 Theories of resource convertibility . . . 3
- 0.4 Phenomenology . . . 3
- 0.5 Monotones and rates (abstract) . . . 4

1 Quantum Resource Theories: General Structure **5**
- 1.1 Definition . . . 5
- 1.2 Consistent QRTs . . . 6
- 1.3 Types of QRTs . . . 7
- 1.4 Resource theories of processes . . . 7
- 1.5 What the quantum setting adds to Chapter 0 . . . 8

2 Resource-Theoretic Tasks **8**
- 2.1 Single-shot convertibility . . . 8
- 2.2 Asymptotic convertibility . . . 9
- 2.3 Catalytic convertibility . . . 10
- 2.4 Convertibility preorder . . . 10
- 2.5 Simulation of non-free operations . . . 10
- 2.6 Erasing resources . . . 11

3 Quantifying Resources **11**
- 3.1 Axioms . . . 11
- 3.2 Distance-based measures . . . 12
- 3.3 Entropic measures . . . 12
- 3.4 Geometric measures . . . 14
- 3.5 Witness-based measures . . . 14

A General Tools **15**
- A.1 Majorization framework . . . 15
- A.2 Convex analysis and SDP . . . 15
- A.3 One-shot and asymptotic tools . . . 15

Further Reading **18**

Chapter 0 Abstract Framework

Idea: *A resource theory is a theory of what can be converted into what, for free. Its minimal mathematical form is a symmetric monoidal category; if one only asks* whether *a conversion exists, it reduces to a commutative preordered monoid.*

0.1 Resource theories as symmetric monoidal categories

Definition 0.1.1 (Resource theory): A *resource theory* is a symmetric (strict) monoidal category $(\mathcal{D}, \circ, \otimes, I)$, where:

- the objects $|\mathcal{D}|$ are the resources;
- the morphisms $\mathcal{D}(A, B)$ are the transformations of A into B that can be implemented at no cost;
- $\circ$ is sequential composition and $\otimes$ is parallel composition;
- I is the void resource, with $A \otimes I = A = I \otimes A$.

Definition 0.1.2 (Free resources): The free resources are $\{A \in |\mathcal{D}| : \mathcal{D}(I, A) \neq \emptyset\}$, i.e. those obtainable from nothing.

Remark 0.1.3: "Resource theory" and "SMC" differ only in interpretation. Strictness is no loss of generality (Mac Lane). Symmetries $A \otimes B \to B \otimes A$ are always free.

0.2 Partitioned process theories

Definition 0.2.1 (Process theory): A process theory is an SMC $(\mathcal{C}, \circ, \otimes, I)$ whose objects are system types and whose morphisms are processes. States are the morphisms $I \to A$.

Definition 0.2.2 (Partitioned process theory): A pair $(\mathcal{C}, \mathcal{C}_{\text{free}})$, where $\mathcal{C}_{\text{free}} \hookrightarrow \mathcal{C}$ is an *all-object-including sub-SMC*. That is, $\mathcal{C}_{\text{free}}$:

- contains all identities and symmetries;
- is closed under $\circ$ and $\otimes$.

The costly processes are $\mathcal{C} \setminus \mathcal{C}_{\text{free}}$. A non-trivial theory requires $\mathcal{C}_{\text{free}} \neq \mathcal{C}$.

Remark 0.2.3: In practice, $\mathcal{C}_{\text{free}}$ is often given by a *generating set* and then closed under $\circ$ and $\otimes$. Checking that the closure is not all of $\mathcal{C}$ is a genuine constraint.

Theorem 0.2.1: Resource theory of states

$\mathbf{S}(\mathcal{C}, \mathcal{C}_{\text{free}})$ is an SMC, hence a resource theory, with:

- objects: the states $s : I \to A$;
- morphisms $s \to t$: free $\xi \in \mathcal{C}_{\text{free}}(A, B)$ with $\xi \circ s = t$, modulo operational equivalence;
- $\otimes$ and I inherited from $\mathcal{C}$.

Definition 0.2.4 (1-comb): A triple (Z, ξ_1, ξ_2) with $\xi_1 \in \mathcal{C}_{\text{free}}(A', A \otimes Z)$ and $\xi_2 \in \mathcal{C}_{\text{free}}(B \otimes Z, B')$. It maps a process $f : A \to B$ to

$$\xi_2 \circ (f \otimes \mathrm{id}_Z) \circ \xi_1 : A' \to B'.$$

Two combs are equivalent if they act identically on all processes, including those acting on larger systems.

Lemma 0.2.5: Any circuit containing a single use of f and otherwise only free processes is a 1-comb applied to f.

Theorem 0.2.2: Resource theory of parallel-combinable processes

$\mathbf{PC}(\mathcal{C}, \mathcal{C}_{\text{free}})$ is an SMC, with:

- objects: the processes $f : A \to B$;
- morphisms $f \to g$: equivalence classes of free 1-combs turning f into g;
- $\otimes$: parallel composition of processes and of combs.

Definition 0.2.6 (n-comb, universal combinability): An *n-comb* is a free circuit $\xi_0, \xi_1, \ldots, \xi_n$ with n holes, filled in some order σ and with side wires Z_i. It turns $(f_1, \ldots, f_n)$ into one process g; this allows *sequential* use of the resources.

- The n-combs form a symmetric multicategory $\mathbf{M}(\mathcal{C}, \mathcal{C}_{\text{free}})$.
- $\mathbf{UC}(\mathcal{C}, \mathcal{C}_{\text{free}})$ has objects $f_1 \boxtimes \cdots \boxtimes f_n$ (finite lists of processes). A morphism to $g_1 \boxtimes \cdots \boxtimes g_m$ is an allocation $\alpha : \{1..n\} \to \{1..m\}$ together with combs $(f_i)_{i \in \alpha^{-1}(j)} \to g_j$.

Remark 0.2.7:

- $\mathbf{S} \subset \mathbf{PC}$: states are processes with trivial input.
- **PC** forbids sequential use of two resources; **UC** allows it.
- Every resource must be consumed somewhere in **UC**, since disposal need not be free.

0.3 Theories of resource convertibility

Definition 0.3.1 (Theory of resource convertibility): A tuple $(\mathcal{R}, +, \succeq, 0)$, where $\succeq$ is a preorder (reflexive and transitive) and $+$ is associative, commutative and unital up to $\simeq$ (with $a \simeq b$ iff $a \succeq b$ and $b \succeq a$). The two structures are compatible:

$$a \succeq b,\ c \succeq d \implies a + c \succeq b + d.$$

That is, a commutative preordered monoid; equivalently, an SMC enriched over $\{0, 1\}$.

Theorem 0.3.1: Partial decategorification

Every resource theory $\mathcal{D}$ defines a theory of resource convertibility via

$$\mathcal{R} = |\mathcal{D}|, \qquad a + b = a \otimes b, \qquad 0 = I, \qquad a \succeq b \iff \mathcal{D}(a, b) \neq \emptyset.$$

Remark 0.3.2:

- $a \simeq b$ does *not* mean $a \cong b$ in $\mathcal{D}$. For convertibility questions, however, $\simeq$ is the right notion of equality.
- The monoid structure is independent of the preorder: the same preorder with $+$ or with $\vee$ gives different theories.
- *Equivalence of theories:* maps $f : \mathcal{R} \to \mathcal{S}$, $g : \mathcal{S} \to \mathcal{R}$ with $gf(a) \simeq a$ and $fg(b) \simeq b$.

0.4 Phenomenology

Definition 0.4.1 (Structural properties): For $\mathcal{R}$:

- *Catalyst:* c is a catalyst for a, b if $a \not\succeq b$ but $a + c \succeq b + c$.
- *Catalysis-free:* $a + c \succeq b + c \Rightarrow a \succeq b$.

- *Non-interacting* (Riesz decomposition): $a \succeq b_1 + b_2 \Rightarrow \exists a_1, a_2$ with $a \simeq a_1 + a_2$, $a_1 \succeq b_1$, $a_2 \succeq b_2$. (With "$\succeq$" in place of "$\simeq$" the condition would be trivial: take $a_i = b_i$.)
- *Quantity-like:* $a_1 + a_2 \simeq b_1 + b_2$ and $a_1 \succeq b_1 \Rightarrow b_2 \succeq a_2$.
- *Quality-like:* $a + a \simeq a$ for all a.
- *Freely disposable:* $a \succeq 0$. *Waste-free:* every a is freely disposable.
- *Riesz interpolation:* $a_i \succeq b_j$ for all i, j (finite families) $\Rightarrow \exists c$ with $a_i \succeq c \succeq b_j$.

Theorem 0.4.1: No catalysis

If $\mathcal{R}$ is non-interacting and quantity-like, then $\mathcal{R}$ is catalysis-free.

Proof. Let $a + c \succeq b + c$. By non-interaction there are a_1, a_2 with $a + c \simeq a_1 + a_2$, $a_1 \succeq b$ and $a_2 \succeq c$. Apply quantity-likeness to $a_2 + a_1 \simeq c + a$ (commutativity) with $a_2 \succeq c$: this gives $a \succeq a_1$. Hence $a \succeq a_1 \succeq b$. □

Proposition 0.4.2 (No-cloning): (i) In every $\mathcal{R}$: $0 \succeq a \Rightarrow a \succeq a + a$.

(ii) If $\mathcal{R}$ is quantity-like: $a \simeq a + a \Rightarrow 0 \succeq a$.

(iii) If $\mathcal{R}$ is quantity-like and waste-free: $a \succeq a + a \iff 0 \succeq a$.

Proof. (i) $a = a + 0 \succeq a + a$ by compatibility.
(ii) Apply quantity-likeness to $a + a \simeq a + 0$ with $a \succeq a$: this gives $0 \succeq a$.
(iii) Waste-free gives $a + a \succeq a + 0 = a$, so $a \succeq a + a$ upgrades to $a \simeq a + a$; then use (ii). The converse is (i). □

Proposition 0.4.3: If $\mathcal{R}$ is quality-like, then $a + a \succeq b \iff a \succeq b \iff a \succeq b + b$. Conversely, this equivalence implies quality-likeness.

Lemma 0.4.4: $\mathcal{R}$ is waste-free iff 0 is a bottom element. In that case $a + b \succeq a$ for all a, b ("the whole is at least each part").

Proof. $a + b \succeq a + 0 = a$, using $b \succeq 0$ and compatibility. □

0.5 Monotones and rates (abstract)

Definition 0.5.1 (Monotone): $M : \mathcal{R} \to \mathbb{R}$ with $a \succeq b \Rightarrow M(a) \geqslant M(b)$. A family (M_i) is *complete* if $a \succeq b \iff M_i(a) \geqslant M_i(b)$ for all i.

- *Additive:* $M(a + b) = M(a) + M(b)$.
- *Supremal:* $M(a + b) = \max\{M(a), M(b)\}$.

Theorem 0.5.1: Complete families exist (Yoneda)

For every $\mathcal{R}$, the family $M_i(a) = 1$ if $a \succeq i$ and 0 otherwise, indexed by $i \in \mathcal{R}$, is complete.

Proof. Each M_i is monotone by transitivity. If $a \not\succeq b$, then $M_b(b) = 1 > 0 = M_b(a)$. □

Remark 0.5.2: The real question is whether a complete family exists that is *smaller* than $\mathcal{R}$, i.e. parametrised by fewer parameters. Additive monotones suit quantity-like theories; supremal monotones suit quality-like ones.

Definition 0.5.3 (Conversion rate): With $n \cdot a = a + \cdots + a$,

$$R(a \to b) := \sup\left\{\tfrac{m}{n} : \; n \cdot a \succeq m \cdot b, \; n, m \in \mathbb{N}_+\right\},$$

and $R := 0$ if no such pair exists. The infimum gives the minimal rate, relevant for disposing of negative resources.

Theorem 0.5.2: Rate bound from additive monotones

Let $M : \mathcal{R} \to \mathbb{R}_{\geqslant 0}$ be a *non-negative* additive monotone with $M(b) > 0$. Then

$$R(a \to b) \leqslant \frac{M(a)}{M(b)}.$$

In particular $R(a \to b) < \infty$.

Proof. $n \cdot a \succeq m \cdot b \Rightarrow nM(a) \geqslant mM(b) \Rightarrow \frac{m}{n} \leqslant \frac{M(a)}{M(b)}$. Take the supremum; if no pair exists, then $R = 0$ and the claim holds since $M(a) \geqslant 0$. □

Remark 0.5.4: Both hypotheses are needed.

- *Sign:* for real-valued M the bound can fail when $R(a \to b) = 0$. On $\mathbb{N}^2$ with the trivial preorder (only $a \succeq a$), $M(x, y) = -x$ is an additive monotone, yet $R((1,0) \to (0,1)) \cdot M(0,1) = 0 > -1 = M(1,0)$.
- $M(b) = 0$*:* then M gives no information about $R(a \to b)$, which may even be ∞.

Remark 0.5.5:

- *Activation:* $k \cdot a \not\succeq k \cdot b$ but $n \cdot a \succeq n \cdot b$ for some $n > k$. This is impossible in quality-like theories.
- The abstract rate is *exact*. Approximate ("epsilonified") rates need extra metric structure (Section 1.5).

Chapter 1 Quantum Resource Theories: General Structure

1.1 Definition

Definition 1.1.1 (QRT): $\mathcal{O}$ assigns to every pair of systems A, B a set $\mathcal{O}(A \to B) \subset \mathcal{Q}(A \to B)$ of CPTP maps. With $\mathcal{F}(\mathcal{H}) := \mathcal{O}(\mathbb{C} \to \mathcal{H})$, the pair $\mathcal{R} = (\mathcal{F}, \mathcal{O})$ is a QRT if:

(i) $\mathrm{id}^A \in \mathcal{O}(A)$;

(ii) $\Phi \in \mathcal{O}(A \to B)$ and $\Lambda \in \mathcal{O}(B \to C) \Rightarrow \Lambda \circ \Phi \in \mathcal{O}(A \to C)$.

Free states: $\mathcal{F}$. Static resources: $\mathcal{S} \setminus \mathcal{F}$. Dynamical resources: $\mathcal{Q} \setminus \mathcal{O}$.

Theorem 1.1.1: Golden rule

$\Phi \in \mathcal{O}(A \to B)$ and $\rho \in \mathcal{F}(A) \Rightarrow \Phi(\rho) \in \mathcal{F}(B)$.

Proof. $\rho \in \mathcal{O}(\mathbb{C} \to A)$, so $\Phi \circ \rho \in \mathcal{O}(\mathbb{C} \to B) = \mathcal{F}(B)$ by (ii). □

Remark 1.1.2: A resource $\sigma \notin \mathcal{F}$ can still be *useful*: there may be $\Phi \in \mathcal{O}$ and $\Lambda \notin \mathcal{O}$ with $\Phi(\rho \otimes \sigma) = \Lambda(\rho)$ for all ρ. This is simulation (Section 2.5).

Definition 1.1.3 (Tensor-product structure): (i) *Completely free:* $\Phi \in \mathcal{O}(A \to B) \Rightarrow \mathrm{id}^C \otimes \Phi \in \mathcal{O}(CA \to CB)$.

(ii) *Appending free states is free:* $\sigma \in \mathcal{F}(B) \Rightarrow (\rho \mapsto \rho \otimes \sigma) \in \mathcal{O}(A \to AB)$.

(iii) *Discarding is free:* $\mathrm{Tr} \in \mathcal{O}(\mathcal{H} \to \mathbb{C})$; this is the only CPTP map to $\mathbb{C}$.

Lemma 1.1.4 (Consequences of a tensor-product structure): (i) $\Phi \otimes \Phi' = (\mathrm{id} \otimes \Phi') \circ (\Phi \otimes \mathrm{id})$ is free, and partial traces are free.

(ii) Replacement channels $X \mapsto \mathrm{Tr}[X]\sigma$ with $\sigma \in \mathcal{F}$ are free, so all free states are interconvertible.

(iii) $\mathcal{F}(A) \otimes \mathcal{F}(B) \subset \mathcal{F}(AB)$, and marginals of free states are free.

Theorem 1.1.2: QRTs as partitioned process theories

Let $\mathcal{R} = (\mathcal{F}, \mathcal{O})$ be a QRT with tensor-product structure in which relabelling of subsystems (the swap) is free. Then $(\mathcal{C}, \mathcal{C}_{\text{free}}) = (\text{quantum channels}, \mathcal{O})$ is a partitioned process theory (Chapter 0). Moreover:

- free states $\mathcal{F}$ = free resources of $\mathbf{S}(\mathcal{C}, \mathcal{O})$;
- state conversion $\rho \xrightarrow{\mathcal{O}} \sigma$ = the preorder of the decategorified $\mathbf{S}$;
- free superchannels = free 1-combs, i.e. $\mathbf{PC}$;
- sequential use of channels = $\mathbf{UC}$;
- discarding free $\Rightarrow$ the theory is waste-free.

Idea. (i)+(ii) of the QRT definition give a subcategory containing all objects. Complete freeness gives closure under $\otimes$, and the free swap gives the symmetries. Waste-freeness: Tr is free, so $\rho \succeq 0$ for every ρ. □

Remark 1.1.5: Without a tensor-product structure (e.g. Bell nonlocality), $\mathcal{O}$ is not closed under $\otimes$: $\rho, \sigma \in \mathcal{F}$ but $\rho \otimes \sigma \notin \mathcal{F}$ (*super-activation*). Such theories are not partitioned process theories in the sense of Chapter 0.

1.2 Consistent QRTs

Two directions. Chapter 0 always starts from $\mathcal{C}_{\text{free}}$. In quantum theory, however, one is often given $\mathcal{F}$ and must choose a compatible $\mathcal{O}$. The two directions behave very differently.

Definition 1.2.1 (Minimal free states): $\mathcal{F}_{\min}(A) := \{\rho : \forall \sigma \in \mathcal{S}(A)\ \exists \Phi \in \mathcal{O}(A),\ \rho = \Phi(\sigma)\}$.

Theorem 1.2.1: $\mathcal{O}$ fixes $\mathcal{F}$

(i) $\mathcal{F}_{\min}(A) \subset \mathcal{F}(A)$.

(ii) If free states are pairwise interconvertible, then $\mathcal{F}(A) = \mathcal{F}_{\min}(A)$.

Proof. (i) Take $\sigma \in \mathcal{F}$. Then $\sigma \to \rho$ for $\rho \in \mathcal{F}_{\min}$, and the golden rule gives $\rho \in \mathcal{F}$.
(ii) For $\sigma \in \mathcal{F}$ and $\rho \in \mathcal{F}_{\min}$: $\omega \to \rho \to \sigma$ for every ω, so $\sigma \in \mathcal{F}_{\min}$. □

Definition 1.2.2 (Classes compatible with $\mathcal{F}$):

- *RNG:* $\mathcal{O}_{\max}(A \to B) = \{\Phi \in \mathcal{Q} : \Phi(\mathcal{F}(A)) \subset \mathcal{F}(B)\}$.
- *k-RNG:* $\mathrm{id}_k \otimes \Phi \in \mathcal{O}_{\max}$. *Completely RNG:* k-RNG for all k; the set is $\mathcal{O}^c_{\max}$.

- *Physically implementable:* compositions of free ancillas, free unitaries, free projective measurements and classical post-processing. The minimal class is $\mathcal{O}_{\min}$.
- *Dually RNG:* $\Phi \in \mathcal{O}_{\max}$ and $\Phi^\dagger(\rho)/\operatorname{Tr}[\Phi^\dagger(\rho)] \in \mathcal{F}$ for all $\rho \in \mathcal{F}$.
- *Stochastically RNG:* a Kraus decomposition with $K_j \rho K_j^\dagger / \operatorname{Tr}[\cdot] \in \mathcal{F}$ for every j and every $\rho \in \mathcal{F}$.
- *ϵ-RG:* $\sup_{\rho\in\mathcal{F}} R_{\text{rob}}(\Phi(\rho)) < \epsilon$; this violates the golden rule only slightly.

Theorem 1.2.2: Hierarchy

- Every compatible $\mathcal{O}$ satisfies $\mathcal{O} \subset \mathcal{O}_{\max}$.
- RNG = 1-RNG $\supset$ 2-RNG $\supset \cdots \supset$ d_A-RNG = CRNG.
- With a tensor-product structure, $\mathcal{O} \subset \mathcal{O}^c_{\max}$.
- $\mathcal{O}_{\min} \subset \mathcal{O}^c_{\max} \subset \mathcal{O}_{\max}$.

1.3 Types of QRTs

Definition 1.3.1 (Convex, affine):

- *Convex:* every $\mathcal{O}(A \to B)$ is convex.
- *Affine:* every CPTP affine combination ($\sum t_j = 1$, $t_j \in \mathbb{R}$) of free maps is free.

Lemma 1.3.2:

- Convex $\mathcal{O} \Rightarrow$ convex $\mathcal{F}$.
- $\mathcal{O}_{\max}$ is convex $\iff$ $\mathcal{F}$ is convex.
- Affine $\mathcal{O} \Rightarrow$ affine $\mathcal{F}$.
- $\mathcal{F}$ affine $\iff$ $\mathcal{O}_{\max}$ affine.

Remark 1.3.3 (Forgetting): The map $\{\rho_i, p_i\} \mapsto \sum p_i \rho_i$ is RNG iff $\mathcal{F}$ is convex. Whether discarding a *classical register*, $\sum p_i \rho_i \otimes |i\rangle\langle i| \mapsto \sum p_i \rho_i$, is free is a separate question, unrelated to convexity.

Definition 1.3.4 (Resource-destroying map): A (possibly non-linear) map Δ with $\Delta(\rho) = \rho$ for $\rho \in \mathcal{F}$ and $\Delta(\rho) \in \mathcal{F}$ for all ρ.

Theorem 1.3.1: Linear resource-destroying maps force affinity

If a linear resource-destroying map exists, then $\mathcal{F}$ is affine, hence convex. Consequently, in a non-convex QRT every resource-destroying map is non-linear.

Proof. For $\sigma = \sum t_j \sigma_j \geqslant 0$ with $\sigma_j \in \mathcal{F}$: $\Delta(\sigma) = \sum t_j \sigma_j = \sigma$, and $\Delta(\sigma) \in \mathcal{F}$. □

Classification. All QRTs $\supset$ convex QRTs $\supset$ affine QRTs $\supset$ QRTs admitting a *linear* resource-destroying map. Non-linear resource-destroying maps can also exist in non-convex theories.

1.4 Resource theories of processes

Definition 1.4.1 (Superchannel, free superchannel): A *superchannel* is a linear map $\Theta : \mathcal{B}(A \to B) \to \mathcal{B}(C \to D)$ that is completely channel-preserving. By Chiribella–D'Ariano–Perinotti, every superchannel has the form

$$\Theta(\Psi) = \Gamma_{\text{post}} \circ (\Psi \otimes \mathrm{id}^E) \circ \Gamma_{\text{pre}}.$$

It is *free* if Γ_{pre} and Γ_{post} are free. This is exactly a free 1-comb, i.e. **PC** of Chapter 0.

Resource calculus. Notation:

- c, q: classical / quantum system;
- $[\cdot], \{\cdot\}$: noiseless / noisy resource;
- $\to$: a channel (dynamical) as opposed to a state (static).

Basic protocols, with $\geqslant$ meaning free simulation:

- teleportation: $[qq] + 2[c \to c] \geqslant [q \to q]$;
- superdense coding: $[qq] + [q \to q] \geqslant 2[c \to c]$;
- cobit $[q \to qq]$: $[q \to qq] \geqslant [qq]$ and $[q \to qq] \geqslant [c \to c]$;
- Harrow's equality: $[qq] + [q \to q] = 2[q \to qq]$.

This is the theory of resource convertibility of **UC**, with the monoid written additively.

1.5 What the quantum setting adds to Chapter 0

Remark 1.5.1: A QRT is a partitioned process theory *with extra structure*:

(i) *Linear/convex structure* on the hom-sets: convexity, affinity, resource-destroying maps, SDP methods.

(ii) *Classical subsystems and flags:* stochastic conversion and strong monotonicity.

(iii) *A metric* (trace distance or fidelity): approximate conversion, ϵ-RG, the asymptotic rates of Chapter 2, which "epsilonify" the exact rates of Section 0.5.

(iv) *The reverse direction* $\mathcal{F} \mapsto \mathcal{O}$: RNG, CRNG, dually RNG.

Each item is a possible enrichment of the abstract framework.

Chapter 2 Resource-Theoretic Tasks

Remark 2.0.1: Abstractly, every task below is a question about the preordered monoid $(\mathcal{R}, \otimes, \succeq, I)$ of Chapter 0. Quantum theory adds two layers of structure: probabilities (stochastic conversion, flags) and a metric (ϵ-approximation).

2.1 Single-shot convertibility

Definition 2.1.1 (Exact conversion): $\rho \xrightarrow{\mathcal{O}} \sigma$ iff $\exists\, \Phi \in \mathcal{O}(A \to B)$ s.t. $\Phi(\rho) = \sigma$.

Definition 2.1.2 (Stochastic conversion): $\rho \xrightarrow{\mathrm{SO}} \sigma$ iff there is a free flagged map $\Phi = \sum_j \Phi_j \otimes |j\rangle\langle j|^X$ with $\Phi_j(\rho)/p_j = \sigma$ for some j, where $p_j = \mathrm{Tr}[\Phi_j(\rho)] > 0$. The optimal success probability is

$$P_\rho^{\max}(\sigma) := \sup\Big\{ \mathrm{Tr}[\Phi_0(\rho)] : \tfrac{\Phi_0(\rho)}{\mathrm{Tr}[\Phi_0(\rho)]} = \sigma,\ \Phi_0 \text{ completable to a free flagged map} \Big\}.$$

Definition 2.1.3 (Approximate conversion): Fix a metric d on states that is contractive under CPTP maps, e.g. the trace distance or the purified distance $P(\rho,\sigma) = \sqrt{1 - F(\rho,\sigma)^2}$. Then $\rho \xrightarrow{\mathcal{O}}_\epsilon \sigma$ iff $\exists\, \sigma' \in B_\epsilon(\sigma)$ with $\rho \xrightarrow{\mathcal{O}} \sigma'$, where $B_\epsilon(\sigma) = \{\sigma' : d(\sigma,\sigma') \leqslant \epsilon\}$.

Remark 2.1.4: A fidelity ball $\{F \geqslant 1-\epsilon\}$ is common in the literature, but $1-F$ is not a metric, so error bounds do not simply add. Using a genuine metric (triangle inequality and contractivity) makes error accumulation transparent; see Section 2.4.

Remark 2.1.5: Exact conversion is often too strict. Stochastic and approximate conversion are its two standard relaxations.

2.2 Asymptotic convertibility

Definition 2.2.1 (Asymptotic rate): Assume a tensor-product structure, so that $\mathcal{O}(A^{\otimes n} \to B^{\otimes m})$ is defined for all n, m. Fix a contractive metric d, e.g. the trace distance. A rate $r \geqslant 0$ is *achievable* for $\rho \to \sigma$ if there is a *sequence* of free maps $\Phi_n \in \mathcal{O}(A^{\otimes n} \to B^{\otimes \lfloor rn \rfloor})$ with

$$\lim_{n\to\infty} d\big(\Phi_n(\rho^{\otimes n}),\ \sigma^{\otimes \lfloor rn \rfloor}\big) = 0.$$

The optimal rate is

$$R(\rho \to \sigma) := \sup\{r :\ r \text{ achievable}\}.$$

Remark 2.2.2: A sequence with vanishing error is essential. Requiring only "for every ϵ some n" would allow $n = 1$: for $r < 1$ the target $\sigma^{\otimes \lfloor r \rfloor}$ is the trivial system, so every $r < 1$ would be achievable for every pair of states.

Lemma 2.2.3 (Basic properties): (i) If free states can be prepared, then $R(\rho \to \sigma) = +\infty$ for all $\sigma \in \mathcal{F}$.

(ii) If $\mathcal{F}$ is closed and discarding is RNG, then $R(\rho \to \sigma) = 0$ for $\rho \in \mathcal{F}$, $\sigma \notin \mathcal{F}$.

(iii) $R(\rho \to \omega) \geqslant R(\rho \to \sigma) \cdot R(\sigma \to \omega)$.

Proof sketch. (i) Prepare $\sigma^{\otimes \lfloor rn \rfloor}$ directly, for any r.

(ii) If $d(\Phi_n(\rho^{\otimes n}), \sigma^{\otimes m_n}) \to 0$ with $m_n \geqslant 1$, discard all but one output copy. This gives free states converging to σ, and closedness of $\mathcal{F}$ gives $\sigma \in \mathcal{F}$.

(iii) Compose achieving sequences Φ_n (rate r_1) and Ψ_m (rate r_2) with $m = \lfloor r_1 n \rfloor$. By contractivity and the triangle inequality the errors add. Discarding a sublinear number of copies adjusts the floors, so every rate below $r_1 r_2$ is achievable. □

Definition 2.2.4 (Reversibility): Restrict to the set $\mathcal{S}_\star$ of states with $R(\rho \to \rho) = 1$; this excludes e.g. freely preparable states, for which $R(\rho \to \rho) = +\infty$. For $\rho, \sigma \in \mathcal{S}_\star$ with $R(\rho \to \sigma), R(\sigma \to \rho) \in (0, \infty)$: $\rho \sim \sigma$ iff $R(\rho \to \sigma) \cdot R(\sigma \to \rho) = 1$.

Lemma 2.2.5: $\sim$ is an equivalence relation on $\mathcal{S}_\star$.

Proof. Reflexivity holds by the definition of $\mathcal{S}_\star$, and symmetry is clear. For transitivity, let $\rho \sim \sigma$ and $\sigma \sim \omega$. By Lemma 2.2.3 (iii),

$$R(\rho \to \omega)\, R(\omega \to \rho) \geqslant R(\rho \to \sigma) R(\sigma \to \omega)\, R(\omega \to \sigma) R(\sigma \to \rho) = 1,$$

and also $R(\rho \to \omega) R(\omega \to \rho) \leqslant R(\rho \to \rho) = 1$. □

Remark 2.2.6:

- Fixing one representative τ_0 in each class, every rate inside the class follows from $R(\rho \to \tau_0)$.
- Distillation rate: $R(\rho \to \tau_0)$. Cost: $R(\tau_0 \to \rho)$.

Definition 2.2.7 (Bound resource): ρ is a bound resource if $R(\tau_0 \to \rho) > 0$ but $R(\rho \to \tau_0) = 0$. This is the extreme form of irreversibility.

2.3 Catalytic convertibility

Definition 2.3.1 (Catalytic convertibility, strict catalysis):

- *Catalytic convertibility:* $\rho \xrightarrow{c\mathcal{O}} \sigma$ iff $\exists\,\omega$ with $\rho \otimes \omega \xrightarrow{\mathcal{O}} \sigma \otimes \omega$. The trivial catalyst is allowed, so $\rho \xrightarrow{\mathcal{O}} \sigma \Rightarrow \rho \xrightarrow{c\mathcal{O}} \sigma$.
- *Strict catalytic advantage:* ω is a catalyst for $\rho \to \sigma$ if, in addition, $\rho \not\xrightarrow{\mathcal{O}} \sigma$.
- *Stochastic version:* $P^{\max}_{\rho\otimes\omega}(\sigma \otimes \omega) > P^{\max}_{\rho}(\sigma)$.

Definition 2.3.2 (Embezzling family): A family $\{\omega_n\}$ with $\omega_n \xrightarrow{\mathcal{O}}_{\epsilon} \sigma \otimes \omega_n$ for all σ, all $\epsilon > 0$ and all sufficiently large n.

Remark 2.3.3: If error is allowed on the catalyst as well, embezzlement trivialises approximate catalytic conversion. Where the error is allowed to sit must therefore be specified.

2.4 Convertibility preorder

Remark 2.4.1:

- $\xrightarrow{\mathcal{O}}$ is a preorder (identity and composition of free maps).
- $\xrightarrow{c\mathcal{O}}$ is a preorder under a tensor-product structure. Reflexivity: trivial catalyst. Transitivity: from $\rho \otimes \omega \to \sigma \otimes \omega$ and $\sigma \otimes \omega' \to \tau \otimes \omega'$, use the catalyst $\omega \otimes \omega'$ and complete freeness:
$$\rho \otimes \omega \otimes \omega' \to \sigma \otimes \omega \otimes \omega' \to \tau \otimes \omega \otimes \omega'.$$
The *strict* catalytic relation is not a preorder: it is never reflexive.
- $\xrightarrow{\mathrm{SO}}$ is a preorder if classically controlled free operations are free (compose the second map conditioned on the success flag of the first).
- The relation "$R(\rho \to \sigma) \geqslant 1$" is reflexive and, by Lemma 2.2.3 (iii), transitive.
- *Error accumulation:* let d be contractive and satisfy the triangle inequality. If $\Phi_1(\rho) = \sigma'$ with $d(\sigma', \sigma) \leqslant \epsilon_1$ and $\Phi_2(\sigma) = \tau'$ with $d(\tau', \tau) \leqslant \epsilon_2$, then
$$d(\Phi_2(\sigma'), \tau) \leqslant d(\Phi_2(\sigma'), \Phi_2(\sigma)) + d(\tau', \tau) \leqslant \epsilon_1 + \epsilon_2.$$
So $\rho \to_{\epsilon_1} \sigma$ and $\sigma \to_{\epsilon_2} \tau$ give $\rho \to_{\epsilon_1+\epsilon_2} \tau$ ("almost transitive").
- Two extremes are uninteresting: almost nothing is comparable, or everything is interconvertible. A meaningful QRT lies in between.

2.5 Simulation of non-free operations

Definition 2.5.1 (Resource-assisted simulation): $\Lambda \notin \mathcal{O}(A \to B)$ is simulated by $\Phi \in \mathcal{O}(AC \to B)$ and $\nu \notin \mathcal{F}(C)$ if $\Phi(\rho \otimes \nu) = \Lambda(\rho)$ for all $\rho \in \mathcal{S}(A)$.

Remark 2.5.2: This converts a static resource into a dynamical one, and it generalises catalysis. Typical questions:

- Which channels are freely simulable at all?
- What is the minimal resource cost of a simulation?

2.6 Erasing resources

Definition 2.6.1 (Erasure by free unitaries): $\rho \mapsto \sum_i p_i U_i \rho U_i^\dagger$ with free unitaries U_i and $\sum_i p_i U_i \rho U_i^\dagger \in \mathcal{F}$.

- Work cost, by Landauer: erasing the classical register costs at least $k_B T \ln 2 \cdot H(p)$ per copy (H in bits), with the register held at temperature T.
- Conservative cost: the minimal number N of unitaries.

Remark 2.6.2 (Catalytic erasure: a model-dependent result): Anshu, Hsieh and Jain (PRL **121**, 190504, 2018) consider a specific catalytic erasure model: free unitaries assisted by a free catalyst, under explicit assumptions on $\mathcal{F}$ and on the set of allowed unitaries. In that model, the optimal asymptotic randomness cost of erasure equals $R_{\mathrm{rel}}^{\infty}$, and the one-shot cost is given by a smoothed max-relative entropy.

Convexity together with a tensor-product structure does *not* suffice. If the only free unitaries are permutations of subsystems, then $\rho^{\otimes n}$ of a pure resource state is invariant under every free unitary and can never be erased, although $R_{\mathrm{rel}}^{\infty}(\rho) > 0$.

Chapter 3 Quantifying Resources

Remark 3.0.1: Abstract monotones (Section 0.5) only respect the preorder. The quantum axioms below add normalisation (A1), compatibility with measurements (A3), with mixing (A4), with $\otimes$ (A5), and with the metric (A6).

3.1 Axioms

Definition 3.1.1 (Resource measure): A function $f : \mathcal{S}(\mathcal{H}) \to \mathbb{R}_{\geqslant 0}$.

Essential axioms.

- **(A1) Vanishing on free states:** $\rho \in \mathcal{F} \Rightarrow f(\rho) = 0$. Faithful if, in addition, $f(\rho) = 0 \Rightarrow \rho \in \mathcal{F}$.
- **(A2) Monotonicity:** $f(\Phi(\rho)) \leqslant f(\rho)$ for all $\Phi \in \mathcal{O}$.

Remark 3.1.2:

- (A2) is the quantitative golden rule.
- If free states are interconvertible, any monotone is constant on $\mathcal{F}$, so (A1) follows after a shift.
- Faithfulness is not required: a task-tailored measure may vanish on useless resources.

Convenient axioms.

- **(A3) Strong monotonicity:** $f(\rho) \geqslant \sum_i p_i f(\sigma_i)$ for free measurements with $\sigma_i = \Phi_i(\rho)/p_i$. Physically, (A3) needs appending and removing classical flags to be free.
- **(A4) Convexity:** $f(\sum_i p_i \rho_i) \leqslant \sum_i p_i f(\rho_i)$.
- **(A5) Subadditivity:** $f(\rho \otimes \sigma) \leqslant f(\rho) + f(\sigma)$; additivity if equality holds. Regularisation: $f^{\infty}(\rho) := \lim_n \frac{1}{n} f(\rho^{\otimes n})$.
- **(A6) Asymptotic continuity:** $|f(\rho) - f(\sigma)| \leqslant K\epsilon \log d + c(\epsilon)$, where $\epsilon = \frac{1}{2}\|\rho - \sigma\|_1$ and $c(\epsilon) \to 0$ as $\epsilon \to 0$.

Theorem 3.1.1: Stochastic bound

If $f \geqslant 0$ satisfies (A3) and $f(\sigma) > 0$, then $P_\rho^{\max}(\sigma) \leqslant f(\rho)/f(\sigma)$.

Idea. $f(\rho) \geqslant p_j f(\sigma)$. □

Theorem 3.1.2: Rate bound

Assume:

- a tensor-product structure, with rates as in Definition 2.2.1;
- $f \geqslant 0$ is a monotone under all multi-copy free maps $\mathcal{O}(A^{\otimes n} \to B^{\otimes m})$;
- f is asymptotically continuous (A6) with respect to the trace distance;
- $f^\infty(\rho)$ and $f^\infty(\sigma)$ exist, and $f^\infty(\sigma) > 0$.

Then $R(\rho \to \sigma) \leqslant f^\infty(\rho)/f^\infty(\sigma)$.

Proof. Let r be achievable with maps Φ_n, put $m_n = \lfloor rn \rfloor$ and $\delta_n := \frac{1}{2}\|\Phi_n(\rho^{\otimes n}) - \sigma^{\otimes m_n}\|_1 \to 0$. By monotonicity and (A6) on $B^{\otimes m_n}$ (dimension $d_B^{m_n}$),

$$f(\rho^{\otimes n}) \geqslant f(\Phi_n(\rho^{\otimes n})) \geqslant f(\sigma^{\otimes m_n}) - K\delta_n\, m_n \log d_B - c(\delta_n).$$

Divide by n and let $n \to \infty$. Using $m_n/n \to r$, $f(\sigma^{\otimes m_n})/m_n \to f^\infty(\sigma)$, $\delta_n \to 0$ and $c(\delta_n)/n \to 0$, this gives $f^\infty(\rho) \geqslant r\, f^\infty(\sigma)$. □

Remark 3.1.3: In QRTs with super-activation, no faithful measure can be subadditive.

3.2 Distance-based measures

Theorem 3.2.1: Distance-based monotones

Let d be contractive under CPTP maps. Then $R_d(\rho) := \inf_{\sigma\in\mathcal{F}} d(\rho,\sigma)$ is a monotone in every QRT.

Proof. Using the golden rule ($\Phi(\sigma) \in \mathcal{F}$),

$$R_d(\Phi(\rho)) \leqslant \inf_{\sigma\in\mathcal{F}} d(\Phi(\rho),\Phi(\sigma)) \leqslant \inf_{\sigma\in\mathcal{F}} d(\rho,\sigma) = R_d(\rho).$$ □

Remark 3.2.1:

- R_d is not automatically strongly monotone.
- It is convex if $\mathcal{F}$ is convex and d is jointly convex.

3.3 Entropic measures

Definition 3.3.1 (Rényi divergences):

- Petz: $D_\alpha(\rho\|\sigma) = \frac{1}{\alpha-1}\log \mathrm{Tr}[\rho^\alpha \sigma^{1-\alpha}]$.
- Sandwiched: $\widetilde{D}_\alpha(\rho\|\sigma) = \frac{1}{\alpha-1}\log \mathrm{Tr}\left[(\sigma^{\frac{1-\alpha}{2\alpha}}\rho\,\sigma^{\frac{1-\alpha}{2\alpha}})^\alpha\right]$.

Limits.

- $\alpha \to 1$: the relative entropy $S(\rho\|\sigma)$.
- $\alpha \to \infty$: $D_{\max}(\rho\|\sigma) = \inf\{\lambda : \rho \leqslant 2^\lambda \sigma\}$.
- $\alpha = 0$: $D_0(\rho\|\sigma) = -\log \mathrm{Tr}[\Pi_\rho \sigma]$.

Data processing holds for D_α with $\alpha \in [0,2]$ and for $\widetilde{D}_\alpha$ with $\alpha \in [\frac{1}{2}, \infty]$.

Definition 3.3.2 (Rényi resource measures):

- Petz: $R_\alpha(\rho) := \inf_{\sigma\in\mathcal{F}} D_\alpha(\rho\|\sigma)$, for $\alpha \in [0,2]$.
- Sandwiched: $\widetilde{R}_\alpha(\rho) := \inf_{\sigma\in\mathcal{F}} \widetilde{D}_\alpha(\rho\|\sigma)$, for $\alpha \in [\frac{1}{2}, \infty]$.
- $R_{\max} := \widetilde{R}_\infty = \inf_{\sigma\in\mathcal{F}} D_{\max}(\cdot\|\sigma)$.
- $R_{\mathrm{rel}} := R_1 = \widetilde{R}_1 = \inf_{\sigma\in\mathcal{F}} S(\cdot\|\sigma)$.

Theorem 3.3.1: Properties of Rényi measures

(i) *Monotonicity:* in every QRT, R_α and $\widetilde{R}_\alpha$ are monotones in their data-processing ranges. This follows from Theorem 3.2.1 and the golden rule.

(ii) *Subadditivity:* if $\mathcal{F}(A)\otimes\mathcal{F}(B) \subset \mathcal{F}(AB)$ (e.g. under a tensor-product structure), then $R(\rho\otimes\sigma) \leqslant R(\rho) + R(\sigma)$, by additivity of the divergences on product states.

(iii) *Convexity* (for convex $\mathcal{F}$): R_{rel} is convex (joint convexity of S), and Petz R_α is convex for $\alpha \in [0,1)$. For $\alpha > 1$, and in particular for $R_{\max}$, one only has *quasi-convexity*: $R(\sum_i p_i\rho_i) \leqslant \max_i R(\rho_i)$.

Proposition 3.3.3 (Strong monotonicity of R_{rel}): Assumptions:

- (F1) the free instruments are those $\{\Phi_i\}$ for which the flagged map $\Phi = \sum_i \Phi_i \otimes |i\rangle\langle i|^X$ is free, with X a classical register;
- (F2) every free state on BX becomes, after dephasing X, a QC state $\sum_i q_i\sigma_i \otimes |i\rangle\langle i|$ with all $\sigma_i \in \mathcal{F}(B)$;
- (F3) dephasing X is free.

Then, with $p_i = \operatorname{Tr}\Phi_i(\rho)$ and $\rho_i = \Phi_i(\rho)/p_i$,

$$R_{\mathrm{rel}}(\rho) \geqslant \sum_i p_i\, R_{\mathrm{rel}}(\rho_i).$$

Proof. By monotonicity, $R_{\mathrm{rel}}(\rho) \geqslant R_{\mathrm{rel}}(\Phi(\rho))$. By (F2), (F3) and data processing under dephasing, the infimum for $\Phi(\rho) = \sum_i p_i\rho_i \otimes |i\rangle\langle i|$ may be taken over QC free states. For these,

$$S\Big(\sum_i p_i\rho_i \otimes |i\rangle\langle i| \,\Big\|\, \sum_i q_i\sigma_i \otimes |i\rangle\langle i|\Big) = \sum_i p_i S(\rho_i\|\sigma_i) + D(p\|q) \geqslant \sum_i p_i R_{\mathrm{rel}}(\rho_i),$$

since $D(p\|q) \geqslant 0$. □

Remark 3.3.4: Under analogous assumptions, Chitambar–Gour (Sec. VI) state strong monotonicity also for the Petz quantities R_α with $\alpha \in (1,2]$; the argument uses concavity of the logarithm in place of the chain rule. This is recorded here without proof.

Remark 3.3.5 ($R_{\max}$ is not convex): Take a qubit with $\mathcal{F} = \{\mathbb{I}/2\}$. Then $R_{\max}(\mathrm{diag}(p, 1-p)) = \log_2(2\max\{p, 1-p\})$. For $\rho_1 = \mathrm{diag}(0.6, 0.4)$ and $\rho_2 = \mathrm{diag}(1, 0)$,

$$R_{\max}\big(\tfrac{1}{2}\rho_1 + \tfrac{1}{2}\rho_2\big) = \log_2 1.6 > \tfrac{1}{2}\big(\log_2 1.2 + \log_2 2\big).$$

Remark 3.3.6:

- R_{rel} is asymptotically continuous if $\mathcal{F}$ is convex and $\mathbb{I}/d \in \mathcal{F}$.
- R^∞_{rel} governs asymptotic conversion (Theorem A.3.3).

Definition 3.3.7 (Reversed relative entropy): $R'_{\mathrm{rel}}(\rho) := \inf_{\sigma\in\mathcal{F}} S(\sigma\|\rho)$. It is additive under a tensor-product structure.

3.4 Geometric measures

Definition 3.4.1:

- $R_{\mathrm{Tr}}(\rho) := \inf_{\sigma\in\mathcal{F}} \frac{1}{2}\|\rho-\sigma\|_1$.
- $R_G(\rho) := \inf_{\sigma\in\mathcal{F}} \left(1-F(\rho,\sigma)^2\right)$.

Theorem 3.4.1: Geometric measures

(i) $R_G = 1-2^{-\widetilde{R}_{1/2}}$, since $\widetilde{D}_{1/2} = -\log F^2$.

(ii) $1-\sqrt{1-R_G} \leqslant R_{\mathrm{Tr}} \leqslant \sqrt{R_G}$.

Remark 3.4.2: If $\mathcal{F}$ is the convex hull of its pure states, then R_G coincides with its convex roof over pure-state decompositions (Streltsov et al.). Combined with free classical flags (assumptions as in the strong-monotonicity proposition), this yields strong monotonicity of R_G.

3.5 Witness-based measures

Definition 3.5.1 (Witness): W Hermitian with $\mathrm{Tr}[W\rho] \geqslant 0$ for all $\rho\in\mathcal{F}$ and $\mathrm{Tr}[W\sigma] < 0$ for some σ.

Theorem 3.5.1: Existence of witnesses

If $\mathcal{F}$ is closed and convex, every resource state has a witness (separating hyperplane).

Definition 3.5.2 (Witness measure): $R_W(\rho) := \sup\{-\mathrm{Tr}[X\rho] : X\in\mathcal{F}^*\cap\mathcal{C}\}$, where $\mathcal{C}$ is

- *closed under the adjoints of free maps*: $\Phi^\dagger(\mathcal{C})\subset\mathcal{C}$; and
- *bounded* and containing 0, e.g. $\mathcal{C} = \{X : -m\mathbb{I}\leqslant X\leqslant n\mathbb{I}\}$. Then $0\in\mathcal{F}^*\cap\mathcal{C}$ gives $R_W\geqslant 0$.

Without the bound, $tW\in\mathcal{F}^*\cap\mathcal{C}$ for all $t>0$, and $R_W=+\infty$ on every resource state. With $\mathcal{C}=\{-m\mathbb{I}\leqslant X\leqslant n\mathbb{I}\}$, closure under adjoints holds because $\Phi^\dagger$ is CP and unital: $n\mathbb{I}-\Phi^\dagger(X) = \Phi^\dagger(n\mathbb{I}-X)\geqslant 0$, and similarly for $X+m\mathbb{I}$.

Theorem 3.5.2: Monotonicity of witness measures

R_W is a monotone and $R_W(\rho)\leqslant m$. It is strongly monotone if $\mathcal{C}$ is closed under classical flags.

Proof of monotonicity. If $X\in\mathcal{F}^*$ and Φ is free, then $\mathrm{Tr}[\Phi^\dagger(X)\rho] = \mathrm{Tr}[X\Phi(\rho)]\geqslant 0$ for all $\rho\in\mathcal{F}$, so $\Phi^\dagger(X)\in\mathcal{F}^*\cap\mathcal{C}$. Hence $-\mathrm{Tr}[X\Phi(\rho)] = -\mathrm{Tr}[\Phi^\dagger(X)\rho]\leqslant R_W(\rho)$. □

Definition 3.5.3 (Robustness): $R^{\mathcal{T}}_{\mathrm{rob}}(\rho) := \inf_{\gamma\in\mathcal{T}}\{s : (\rho+s\gamma)/(1+s)\in\mathcal{F}\}$.

- $\mathcal{T}=\mathcal{F}$: absolute robustness.
- $\mathcal{T}=\mathcal{S}$: global robustness.
- $\mathcal{T}=\{\mathbb{I}/d\}$: random robustness.

Theorem 3.5.3: Log-robustness

$R_{\max}(\rho) = \log(1+R_{\mathrm{rob}}(\rho))$, with R_{rob} the global robustness.

Definition 3.5.4 (Resource rank): If the pure free states span $\mathcal{H}$, then

- $R_{\mathrm{rk}}(\psi) :=$ the minimal number of pure free states in a superposition decomposition of ψ;
- for mixed states, take the convex roof.

Appendix A General Tools

Technical tools used in the further reading; not part of the core framework.

A.1 Majorization framework

Definition A.1.1 (Majorization): $q \prec p$ iff $\sum_{i\leqslant k} q_i^{\downarrow} \leqslant \sum_{i\leqslant k} p_i^{\downarrow}$ for all k, with equality at $k = n$.

Theorem A.1.1: Hardy–Littlewood–Pólya, Birkhoff

$q \prec p \iff q = Dp$ for some doubly stochastic $D \iff q \in \mathrm{conv}\{\pi p\}$.

Definition A.1.2 (Relative majorization): $(p', q') \prec_r (p, q)$ iff there is a stochastic M with $p' = Mp$ and $q' = Mq$.

Theorem A.1.2: Blackwell

$(p', q') \prec_r (p, q) \iff T(p', q') \subseteq T(p, q)$, i.e. iff the Lorenz curve of (p', q') never lies above that of (p, q).

Definition A.1.3 (Quantum majorization): $\sigma^{AB} \prec_q \rho^{AB}$ iff $\sigma = (\mathrm{id}\otimes\mathcal{E})(\rho)$ for some channel $\mathcal{E}$, possibly constrained (e.g. G-covariant).

Remark A.1.4: Majorization-type preorders give complete sets of monotones for pure or quasi-classical conversion problems.

A.2 Convex analysis and SDP

Theorem A.2.1: Weak membership

If $\mathcal{F}$ is closed and convex, then $\rho \in \mathcal{F} \iff \mathrm{Tr}[W\rho] \geqslant 0$ for all $W \in \mathcal{F}^*$.

Theorem A.2.2: Conversion as a conic program

Let $\mathcal{C} = \{J_\Phi^{AB} : \Phi \in \mathcal{O}(A \to B)\}$, with $J_\Phi^{AB} = (\mathrm{id}_A \otimes \Phi)(\phi^+)$, be closed and convex. Then $\rho \xrightarrow{\mathcal{O}} \sigma$ iff there is J with

$$\mathrm{Tr}_B J^{AB} = \mathbb{I}_A, \qquad \mathrm{Tr}_A[J^{AB}(\rho^T \otimes \mathbb{I}_B)] = \sigma, \qquad \mathrm{Tr}[JW] \geqslant 0 \ \ \forall W \in \mathcal{C}^*.$$

Remark A.2.1: For affine QRTs this is an SDP. In general it can be hard.

A.3 One-shot and asymptotic tools

Definition A.3.1 (Smooth max-resource): $R_{\max}^\epsilon(\rho) := \inf_{\hat\rho\in B_\epsilon(\rho)} R_{\max}(\hat\rho)$, with B_ϵ a purified-distance ball. In suitable operational models (e.g. RNG or δ-RG operations) it quantifies the one-shot resource *cost*; its operational meaning is model dependent.

Definition A.3.2 (Hypothesis-testing relative entropy):

- $$D_H^\epsilon(\rho\|\sigma) := \sup_{\substack{0\leqslant M\leqslant\mathbb{I} \\ \mathrm{Tr}[(\mathbb{I}-M)\rho]\leqslant\epsilon}} -\log \mathrm{Tr}[M\sigma].$$
- $R_H^\epsilon(\rho) := \inf_{\sigma\in\mathcal{F}} D_H^\epsilon(\rho\|\sigma)$. In suitable models (e.g. RNG operations) it quantifies the one-shot *yield* (distillation).

Theorem A.3.1: Quantum Stein lemma

$\lim_n \frac{1}{n} D_H^\epsilon(\rho^{\otimes n} \| \sigma^{\otimes n}) = S(\rho\|\sigma)$ for all $\epsilon \in (0,1)$.

Setting. A family $\{\mathcal{F}_n\}_n$, with $\mathcal{F}_n \subset \mathcal{S}(\mathcal{H}^{\otimes n})$, satisfying the Brandão–Plenio axioms:

(i) each $\mathcal{F}_n$ is closed and convex;

(ii) $\mathcal{F}_n$ contains $\sigma_0^{\otimes n}$ for some full-rank σ_0;

(iii) closure under partial traces: $\mathrm{Tr}_n \mathcal{F}_n \subset \mathcal{F}_{n-1}$;

(iv) closure under tensor products: $\mathcal{F}_n \otimes \mathcal{F}_m \subset \mathcal{F}_{n+m}$;

(v) closure under permutations of the n copies.

Then $R_{\mathrm{rel}}^\infty(\rho) := \lim_n \frac{1}{n} \inf_{\sigma \in \mathcal{F}_n} S(\rho^{\otimes n}\|\sigma)$ exists.

Theorem A.3.2: Generalised quantum Stein lemma

Under axioms (i)–(v), for every $\epsilon \in (0,1)$ (with R_H^ϵ computed relative to $\mathcal{F}_n$):

- $\lim_n \frac{1}{n} R_H^\epsilon(\rho^{\otimes n}) = R_{\mathrm{rel}}^\infty(\rho)$;
- $\lim_{\epsilon\to 0} \lim_n \frac{1}{n} R_{\max}^\epsilon(\rho^{\otimes n}) = R_{\mathrm{rel}}^\infty(\rho)$.

Remark A.3.3 (Status of the proof):

- The lemma was stated by Brandão–Plenio (2010).
- A gap in that proof was identified by Berta et al. (Quantum, 2023), which left the lemma and its consequences unproven.
- Proofs were then given independently by Hayashi–Yamasaki (Nature Physics, 2025, under weaker assumptions; extended to classical-quantum channels) and by Lami (IEEE Trans. Inf. Theory, 2025).

See Further Reading B.

Theorem A.3.3: Asymptotic reversibility (Brandão–Gour)

Assume axioms (i)–(v). Let the free operations be *asymptotically RNG*: maps Φ_n that are ϵ_n-RG with respect to $\{\mathcal{F}_n\}$, with $\epsilon_n \to 0$. Define rates as in Definition 2.2.1, with Φ_n asymptotically RNG instead of free. Then, for all ρ, σ with $R_{\mathrm{rel}}^\infty(\rho), R_{\mathrm{rel}}^\infty(\sigma) \in (0,\infty)$,

$$R(\rho \to \sigma) = \frac{R_{\mathrm{rel}}^\infty(\rho)}{R_{\mathrm{rel}}^\infty(\sigma)}.$$

In particular, all such states are reversible. The proof relies on the generalised Stein lemma, so this theorem stands with its recent proofs.

Corollary A.3.4 (Uniqueness): In the setting of Theorem A.3.3, let f be a monotone under the asymptotically RNG operations that is asymptotically continuous, whose regularisation f^∞ exists, and with $f^\infty(\rho), f^\infty(\sigma) \in (0,\infty)$. Then

$$\frac{f^\infty(\rho)}{f^\infty(\sigma)} = \frac{R_{\mathrm{rel}}^\infty(\rho)}{R_{\mathrm{rel}}^\infty(\sigma)}.$$

So R_{rel}^∞ is the unique asymptotic measure up to normalisation.

Proof. The rate bound (Theorem 3.1.2) applied in both directions gives

$$R(\rho \to \sigma) \leqslant \frac{f^\infty(\rho)}{f^\infty(\sigma)} \leqslant \frac{1}{R(\sigma \to \rho)}.$$

By Theorem A.3.3, both ends equal $R^\infty_{\mathrm{rel}}(\rho)/R^\infty_{\mathrm{rel}}(\sigma)$. □

Further Reading

These notes are meant as the minimal background for the readings below. For each resource theory, the relevant chapters are indicated as prerequisites.

A. The abstract framework (Chapter 0)

- B. Coecke, T. Fritz, R. W. Spekkens, *A mathematical theory of resources*, Inf. Comput. **250**, 59 (2016). arXiv:1409.5531
- T. Fritz, *Resource convertibility and ordered commutative monoids*, Math. Struct. Comp. Sci. **27**, 850 (2017). arXiv:1504.03661
- T. Gonda, R. W. Spekkens, *Monotones in general resource theories*, Compositionality (2023). arXiv:1912.07085

B. General quantum resource theories (Chapters 1–3, Appendix A)

- E. Chitambar, G. Gour, *Quantum resource theories*, Rev. Mod. Phys. **91**, 025001 (2019). arXiv:1806.06107
- F. G. S. L. Brandão, G. Gour, *Reversible framework for quantum resource theories*, Phys. Rev. Lett. **115**, 070503 (2015). arXiv:1502.03149
- Z.-W. Liu, X. Hu, S. Lloyd, *Resource destroying maps*, Phys. Rev. Lett. **118**, 060502 (2017). arXiv:1606.03723
- R. Takagi, B. Regula, *General resource theories in quantum mechanics and beyond: operational characterization via discrimination tasks*, Phys. Rev. X **9**, 031053 (2019). arXiv:1901.08127
- M. Berta, F. G. S. L. Brandão, G. Gour, L. Lami, M. B. Plenio, B. Regula, M. Tomamichel, *On a gap in the proof of the generalised quantum Stein's lemma and its consequences for the reversibility of quantum resources*, Quantum **7**, 1103 (2023). arXiv:2205.02813
- M. Hayashi, H. Yamasaki, *The generalized quantum Stein's lemma and the second law of quantum resource theories*, Nature Physics (2025), doi:10.1038/s41567-025-03047-9. arXiv:2408.02722
- L. Lami, *A solution of the generalised quantum Stein's lemma*, IEEE Trans. Inf. Theory **71**, 4454 (2025). arXiv:2408.06410
- A. Anshu, M.-H. Hsieh, R. Jain, *Quantifying resources in general resource theory with catalysts*, Phys. Rev. Lett. **121**, 190504 (2018). arXiv:1708.00381
- G. Gour, A. Winter, *How to quantify a dynamical quantum resource*, Phys. Rev. Lett. **123**, 150401 (2019). arXiv:1906.03517
- I. Devetak, A. W. Harrow, A. Winter, *A resource framework for quantum Shannon theory*, IEEE Trans. Inf. Theory **54**, 4587 (2008). arXiv:quant-ph/0512015

C. Resource theories of specific phenomena

Entanglement. *Prerequisites: Ch. 1.1–1.2, 2, 3.*

- R. Horodecki, P. Horodecki, M. Horodecki, K. Horodecki, *Quantum entanglement*, Rev. Mod. Phys. **81**, 865 (2009). arXiv:quant-ph/0702225
- M. B. Plenio, S. Virmani, *An introduction to entanglement measures*, Quant. Inf. Comput. **7**, 1 (2007). arXiv:quant-ph/0504163

Quantum reference frames and asymmetry. *Prerequisites: Ch. 0.2, 1.3 (resource-destroying maps), 2.2, App. A.3 (asymmetry is non-extensive: $R^{\infty}_{\mathrm{rel}} = 0$, so the Brandão–Gour theorem does not apply).*

- S. D. Bartlett, T. Rudolph, R. W. Spekkens, *Reference frames, superselection rules, and quantum information*, Rev. Mod. Phys. **79**, 555 (2007). arXiv:quant-ph/0610030
- G. Gour, R. W. Spekkens, *The resource theory of quantum reference frames: manipulations and monotones*, New J. Phys. **10**, 033023 (2008). arXiv:0711.0043
- I. Marvian, R. W. Spekkens, *The theory of manipulations of pure state asymmetry*, New J. Phys. **15**, 033001 (2013). arXiv:1104.0018
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Coherence and imaginarity. *Prerequisites: Ch. 1.2 (classes of free operations), 1.3 (affine theories), 3.*

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